

Giovanni Pierobon

Email: tanogp@protonmail.com ◦ GitHub: github.com/gpierobon ◦ Website: gpierobon.github.io
Sydney/Ballina, NSW, Australia

Languages: English (fluent), Spanish (fluent), Italian (native), German (basic)

PROFESSIONAL EXPERIENCE

Research Associate, The University of New South Wales Sydney, 2024 – Current

- Numerical modeling of nuclear magnetic resonance (NMR) systems and implications for cosmic relics and fundamental physics.
- Enhancing the experimental sensitivity of dark matter searches (e.g., microwave cavities) through more accurate modeling of the underlying spectral signatures.
- Lecturing and presenting at leading international workshops and conferences, as well as invited seminars.

PhD researcher, The University of New South Wales Sydney, 2020 – 2023

- Leading the first in-depth numerical treatment of sub-galactic structure of dark matter (axion) by means of high-resolution hybrid simulations (lattice+particle+Monte Carlo).
- Published 7 papers in high-impact factors journals in theoretical physics.

Teaching Assistant, The University of New South Wales Sydney, 2020 – 2024

- Academic tutoring of advanced course in the Bachelor degree within the School of Physics.
- Marking and moderating first year general physics introductory courses.

TECHNICAL EXPERTISE

- **Quantum modeling and simulation**
 - Open quantum systems: master equations, Lindblad formalism, NMR experimental modeling
 - Time evolution solvers, simulation frameworks (QuTiP), mean-field methods
 - Visualization tools: Bloch spheres, density matrices
- **Data Analysis & Statistical Methods**
 - NumPy, SciPy, Pandas, Scikit-learn, Jupyter
 - Markov Chain Monte Carlo, power spectra & correlation function
 - Signal processing (Fourier transforms, windowing, filtering) and noise modeling
 - Visualisation: Matplotlib, Mathematica, volume rendering (yt)
- **High-performance computing**
 - Particle and lattice simulations (finite difference, spectral methods)
 - Parallelisation and acceleration: MPI, OpenMP, CUDA
 - HDF5 I/O, PBS scheduling (CentOS/RHEL), profiling
- **Core:** Python, C, C++, Bash, Linux, shell scripting
- **Tools:** Git, GitHub workflows, CMake, Make, Docker

SOFT SKILLS

- Confident with public speaking
- Balanced between detail-oriented & big-picture
- Self-motivated, autonomous & team-player
- Adaptable & receptive to change

EDUCATION

Ph.D.	The University of New South Wales	Sydney, 2020-2023
M.Sc.	University of Padua	Padua, 2017-2019
M.Sc.	University of Heidelberg (exchange year)	Heidelberg, 2018-2019
B.Sc.	University of Padua	Padua, 2014-2017

Certifications

NVIDIA Deep Learning Institute	Fundamentals of Accelerated Computing with CUDA Python	2023
National Computational Infrastructure	Introduction to OpenMP	2023
Intersect Australia	Linear regression, Classification	2023

ACADEMIC PUBLICATIONS

A list of academic works can be found on the inspire database <https://inspirehep.net/authors/1774791>

CODING PROJECTS

I lead various projects and maintain several sizable codebases. My GitHub contains a list of my repositories <https://github.com/gpierothon>.

INVITED TALKS AND SEMINARS

○ Workshop, Institute for Nuclear Theory	Seattle, Dec. 2025
○ Seminar, Korea Institute for Advanced Study	Seoul, Nov. 2025
○ Seminar, University of Zaragoza	Zaragoza, Sep. 2025
○ Seminar, University of Bielefeld	Bielefeld, Sep. 2025
○ Seminar, Sydney CPPC group	Sydney, Apr. 2025
○ Conference, Cosmic Wispers General Meeting	Istanbul, Sep. 2024
○ Workshop, Advancements in Axion Physics	Shanghai, Oct. 2023
○ Workshop, Galileo Galilei Institute for Theoretical Physics	Florence, May 2023
○ Seminar, University of Zaragoza	Zaragoza, Mar. 2023
○ Conference, Cosmic Wispers kick-off meeting	Rome, Feb. 2023
○ Seminar, UCLA	Los Angeles, Jan. 2023
○ AIP NSW Postgraduate Awards	Sydney, Nov. 2022

TEACHING EXPERIENCE AND SUPERVISION

School of Physics (UNSW Sydney)

○ PHYS3115: Particle physics and the early Universe	2020, 2022
○ PHYS2113: Classical mechanics and special relativity	2021
○ PHYS1160: Introduction to astronomy	2020-2023
○ Special guest for PHYS3116 lecture (Galaxies and Cosmology)	2024

National Computational Infrastructure and Intersect Australia

○ Guest lecture: Running and Analyzing Cosmological Simulations	2024
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I have co-supervised 1 PhD student and 3 Honours (Master) projects:

○ Sean Bryan, Simulations of axion string networks	2021
○ Viraj Ekanayakage Don, Tidal disruption of fuzzy dark matter	2022
○ Juan Minguez-Laguna, Axion simulations with Schrödinger-Poisson	2023
○ Yeray Garcia del Castillo (PhD)	2024-

AWARDS

NCMAS and UNSW compute allocation National Computational Infrastructure	2021-2025
Postgraduate Award Nomination Australian Institute of Physics	2022
University International Postgraduate Award (UIPA) The University of New South Wales	2019